Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

PROBLEM STATEMENT:

Your organization wants to regularly import a list of new employees with their details, such as Name, Email, Department, and Manager, from an external system. Once imported, when an Incident is assigned to an employee, the Incident form should automatically show the manager's email of the employee's department.

OBJECTIVE:

- Import employee records from an external CSV file into ServiceNow
- Automatically map relationships between employees, departments, and managers using dot-walking
- Create a staging table and transform map for structured data loading • Use Mapping Assist for field-level mapping between source and target tables • Configure the Incident form to display dot-walked fields (Department, Email, Manager) • Verify imported data by testing via the Incident form

SKILLS:

Using Import Sets & Dot Walking

TASK INITIATION

MILESTONE 1: Import Data from Data Source

◆ ACTIVITY 1: Create Records in the Sheet

Steps:

1. Open a spreadsheet (Excel or Google Sheets)
2. Add the following column headers: name, e-mail id, phone number, department, manager
3. Fill in employee data rows under each column
4. Rename the sheet tab to: employee data
5. Download / Save the file in CSV format

◆ ACTIVITY 2: Load Data into ServiceNow Import Set

Steps:

6. Login to ServiceNow Developer instance
7. Click Start Building on the homepage
8. Go to All → Search: Import Sets → Click Load Data under Import Sets
9. Fill the following details:

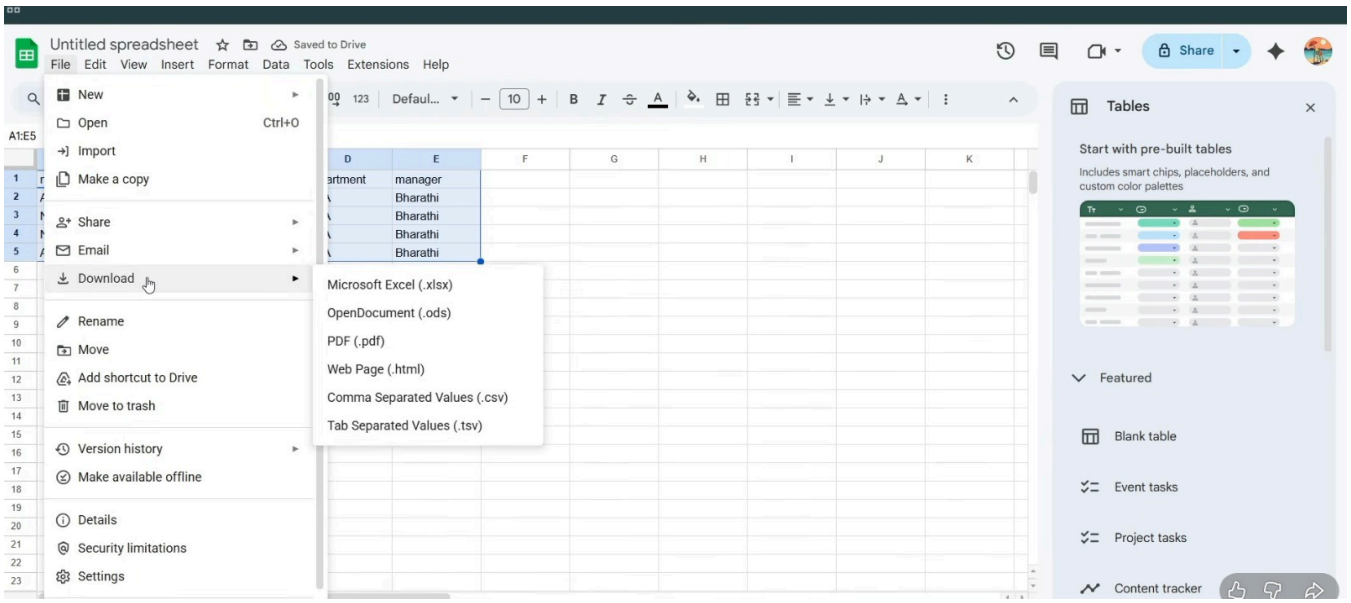
Fill:

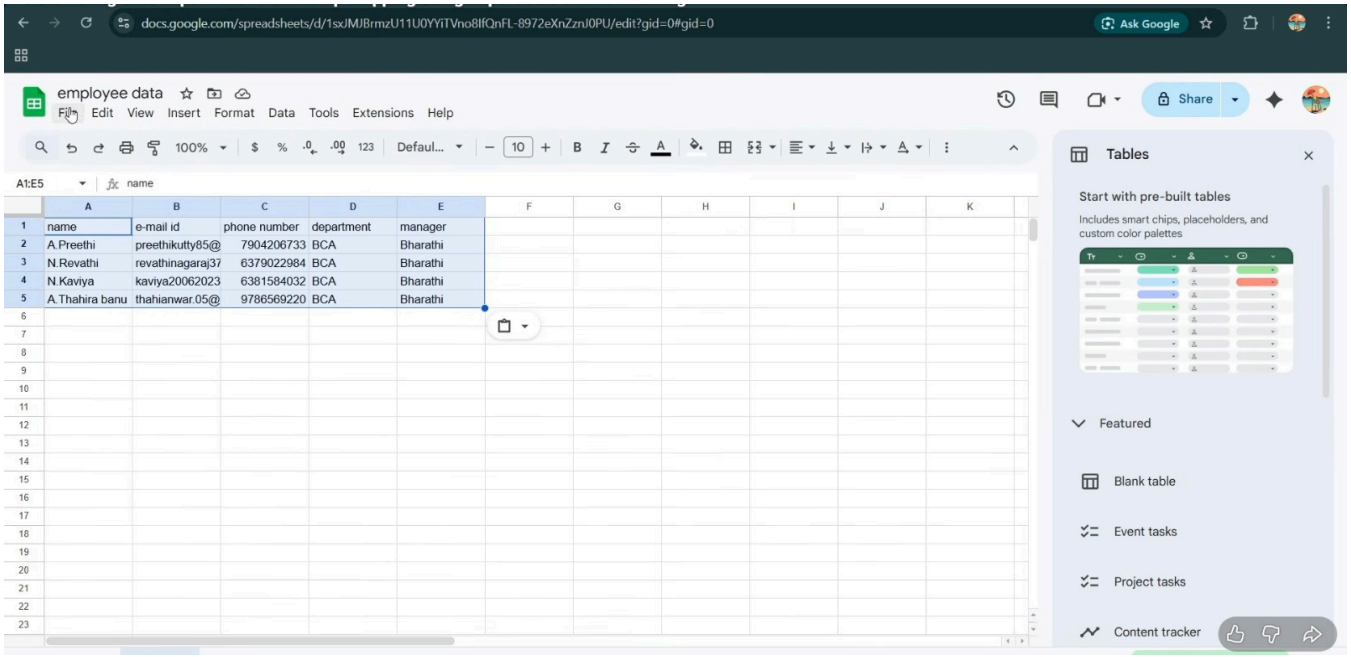
- **Label:** import employee
- **File:** Upload the CSV file created above

10. Click Submit

Result: ImportProcessor runs and data is loaded into the staging table

u_import_employee.





MILESTONE 2: Create Transform Map

◆ ACTIVITY 1: Creating Transform Map

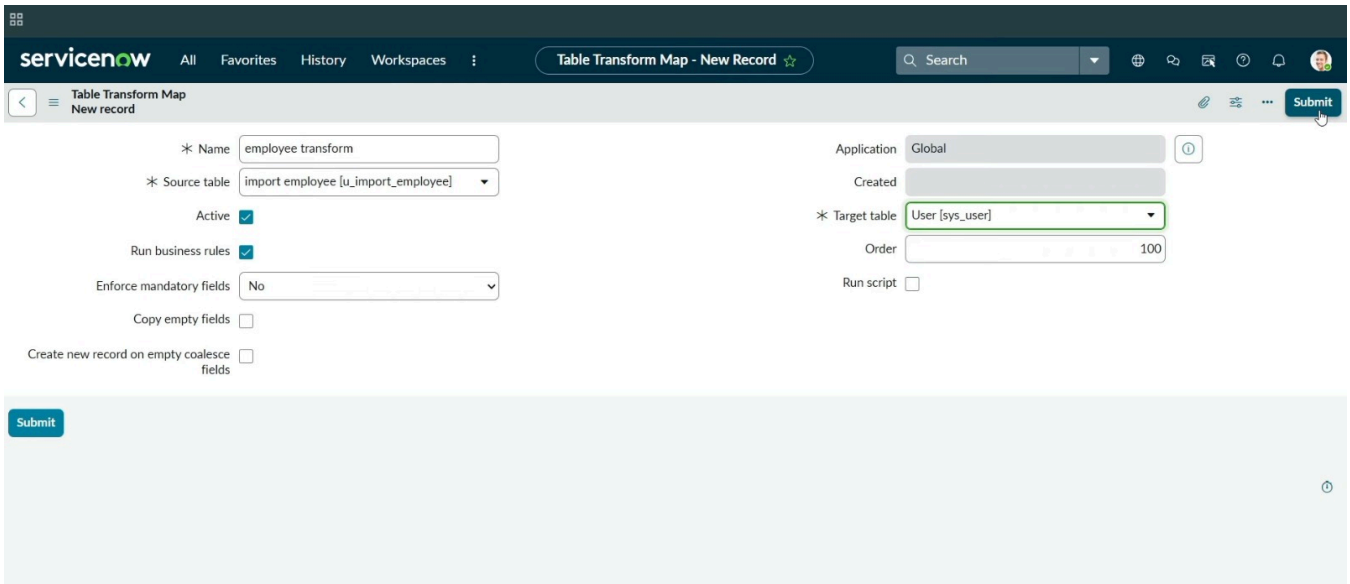
Steps:

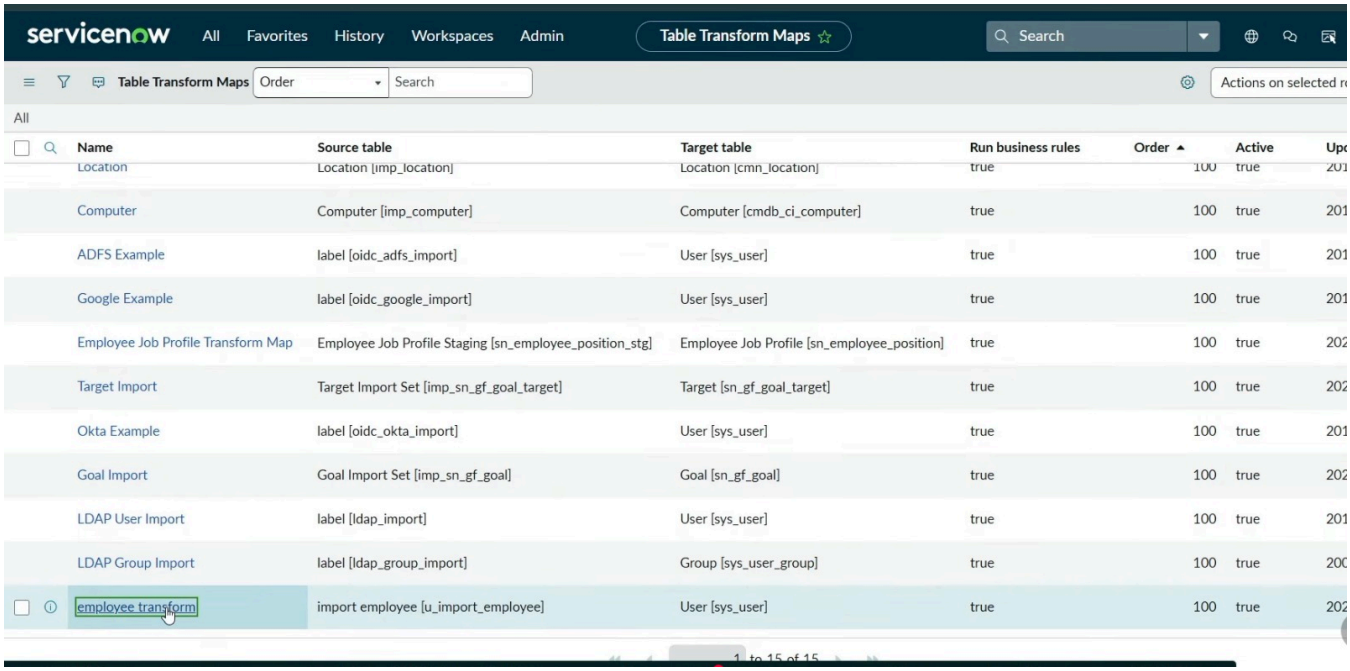
11. Go to All → Search: Transform Map → Click Create Transform Map
12. Fill the following:

Fill:

- **Name:** employee transform
- **Source Table:** import employee [u_import_employee]
- **Target Table:** User [sys_user]

13. Click Submit

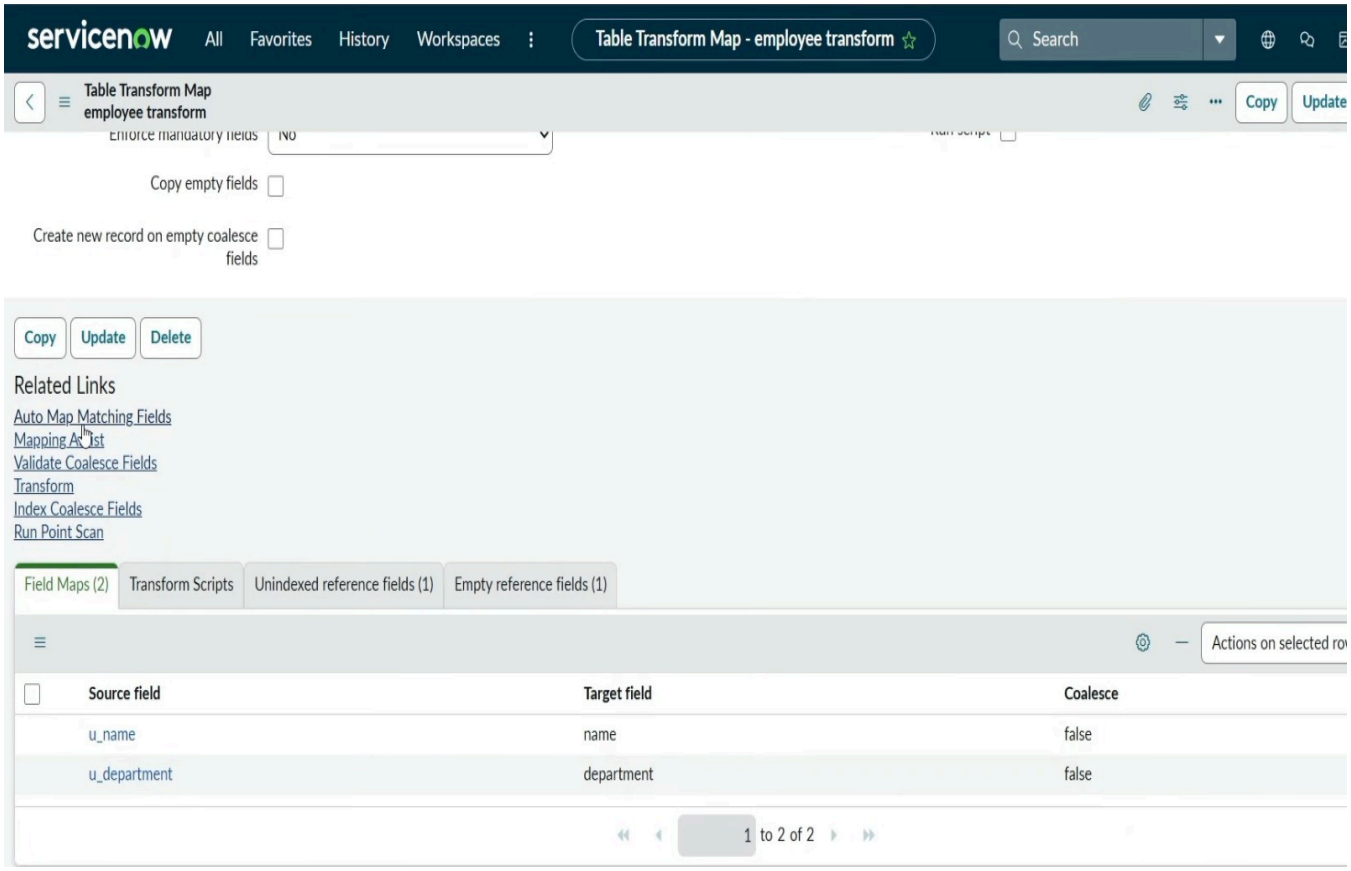


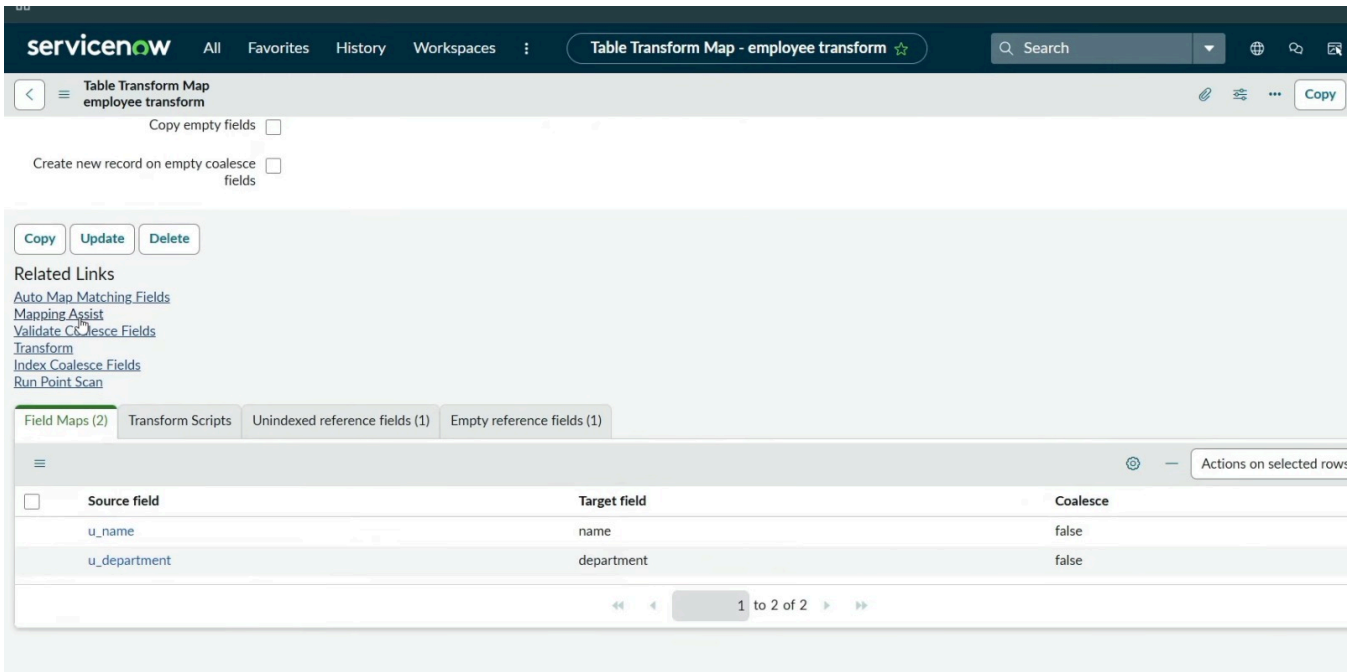


◆ ACTIVITY 2: Auto Map & Mapping Assist

Steps:

14. Scroll down and click employee transform to open it
15. Scroll down and click Auto Map Matching Fields
16. The auto-matched fields will be shown
17. Click Mapping Assist link
18. From Source: import employee — select fields: name, e-mail id, phone number, department, manager
19. Map them to corresponding Target: User fields
20. Click Save



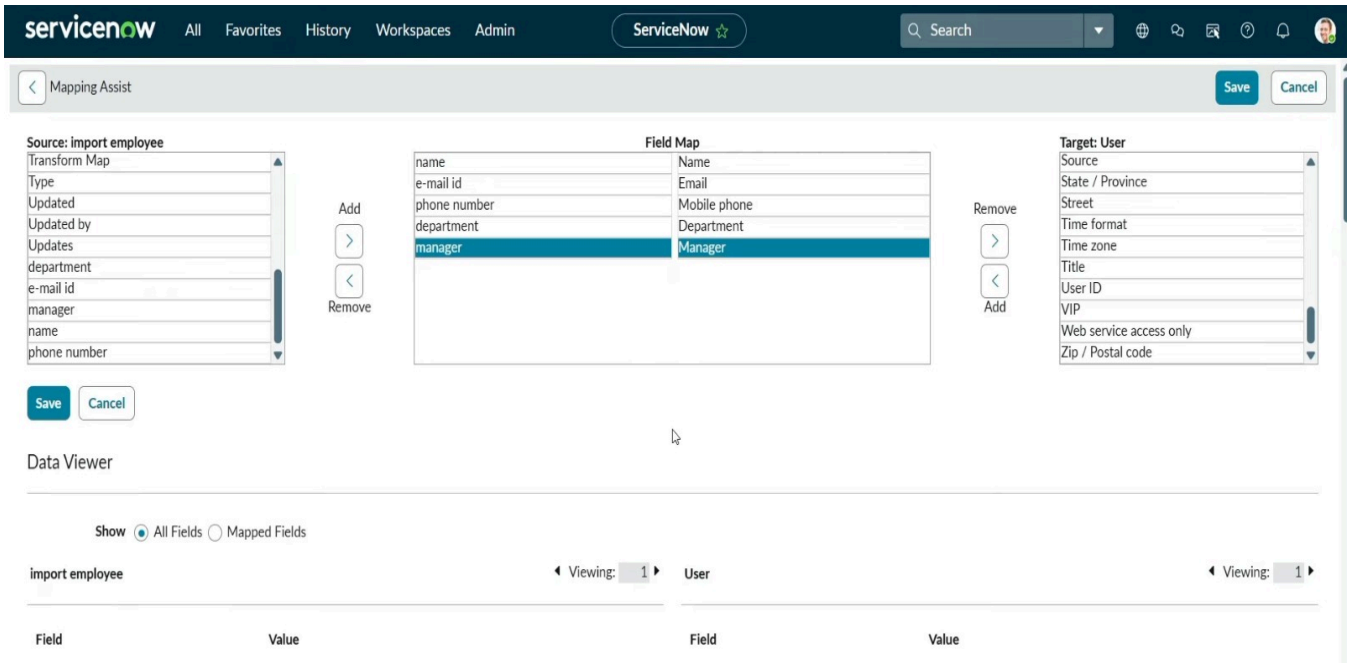


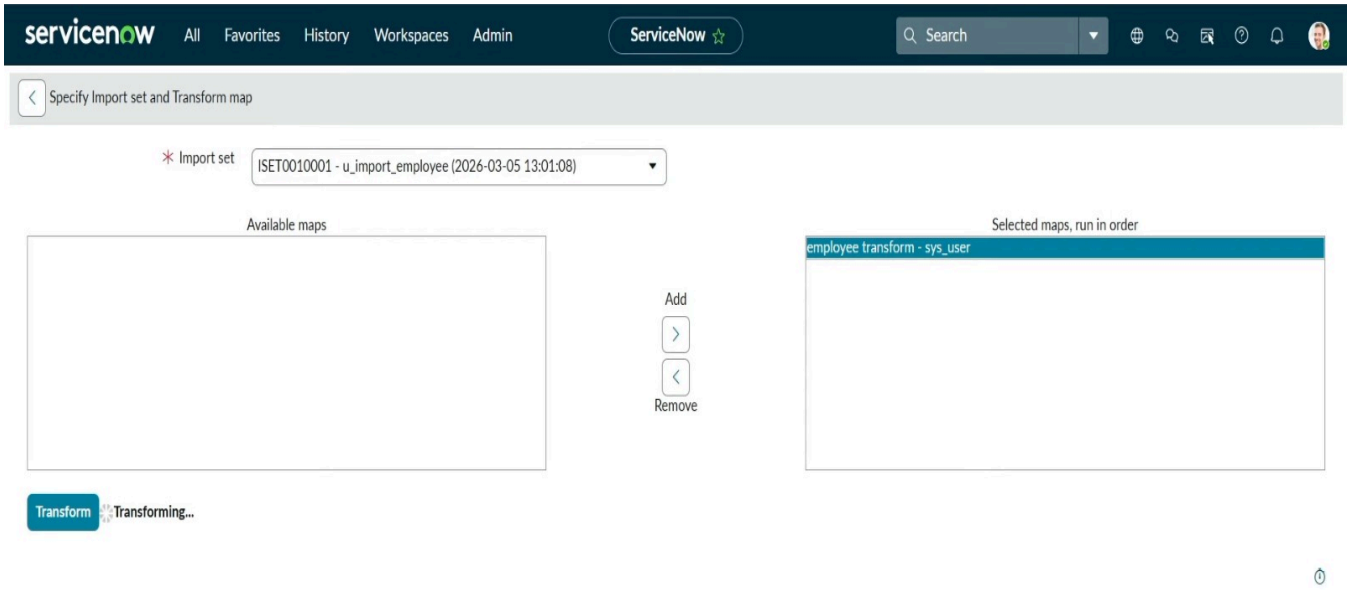
MILESTONE 3: Run Transform

◆ ACTIVITY 1: Execute Transform

Steps:

21. Click Transform link inside the Transform Map
22. Import set: ISET0010001 - u_import_employee is selected
23. Ensure employee transform - sys_user is in Selected maps
24. Click Transform button





Result:

- State: Complete
- Completion code: Success
- Message: Transformation complete

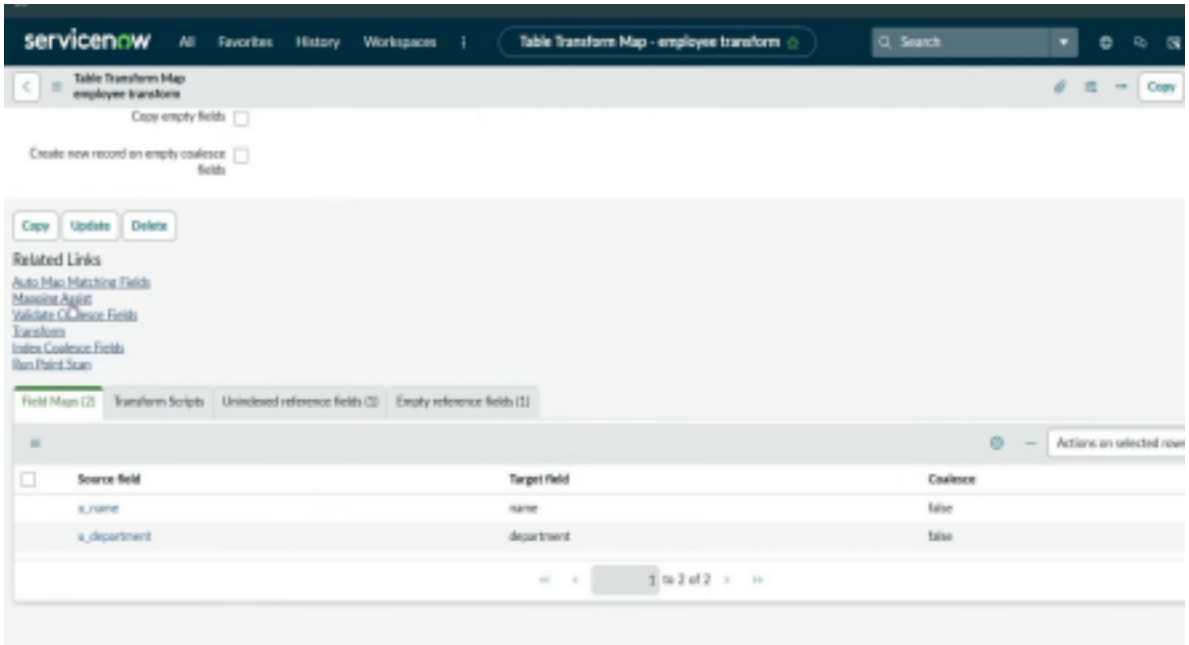
MILESTONE 4: Dot Walking – Configure Incident Form

Dot Walking allows traversal of related table fields and displays them directly on a form. In this task, we configure the Incident form to show the assigned employee's Department, Email, and Manager using dot-walked fields from the User table.

◆ ACTIVITY 1: Configure Incident Form Layout

Steps:

25. Go to All → Search: Incident → Click Create New under Incident
26. On the Incident form, click Configure → Form Layout
27. In the Available column, find Assigned to — expand it
28. Select: Assigned to.Department, Assigned to.Email, Assigned to.Manager
29. Move selected fields to the Selected column
30. Click Save



MILESTONE 5: Testing

◆ ACTIVITY 1: Verify Imported Data in User Table

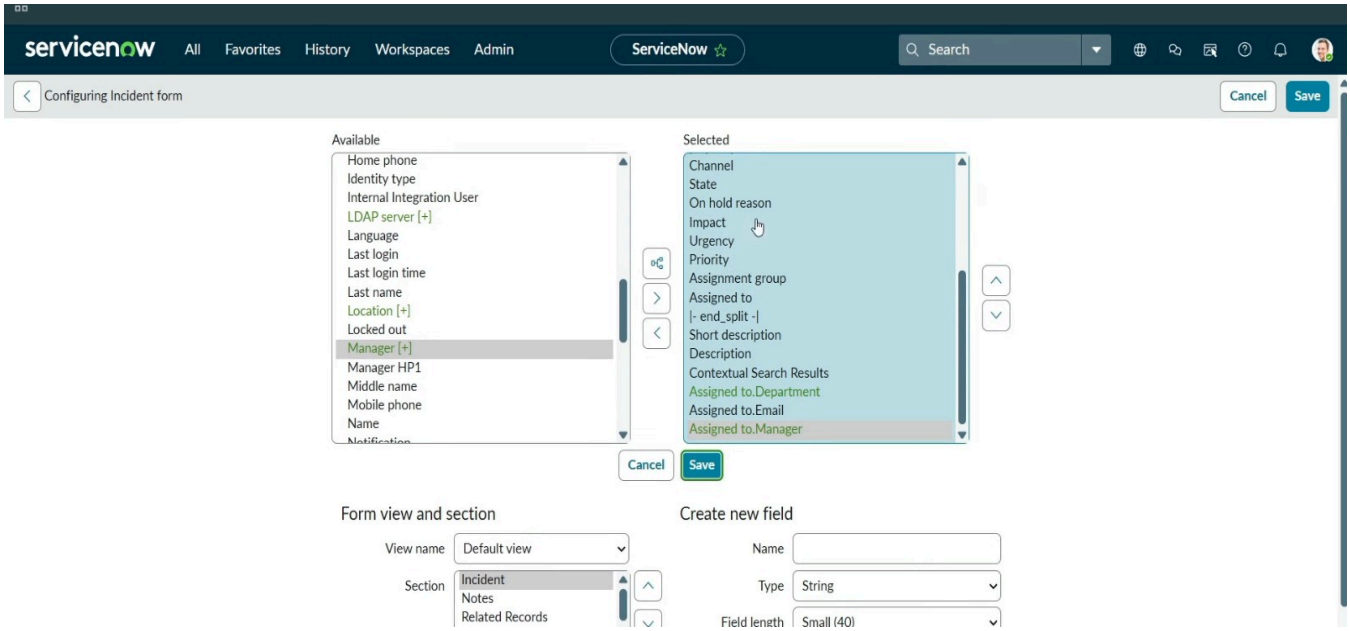
Steps:

31. Go to All → Search: User → Select User under User Administration
32. Verify all imported employee records appear correctly
33. Add User ID for each record and click Update

◆ ACTIVITY 2: Test Dot Walking on Incident Form

Test Case:

34. Go to All → Incident → Create New
35. In the Department field, click search icon — departments popup appears
36. Select a department (e.g., BCA)
37. In the Manager field, type partial name (e.g., bha)
38. System auto-suggests matching manager records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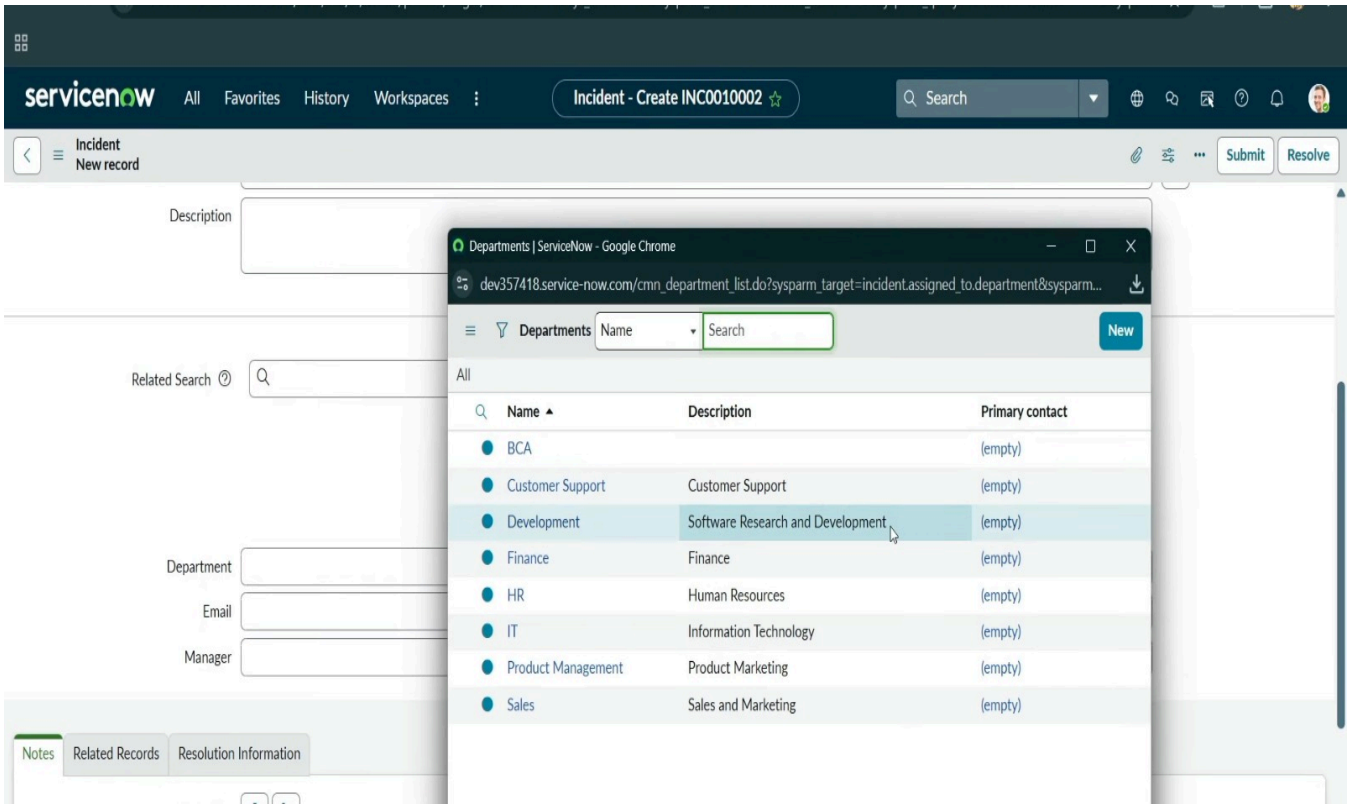
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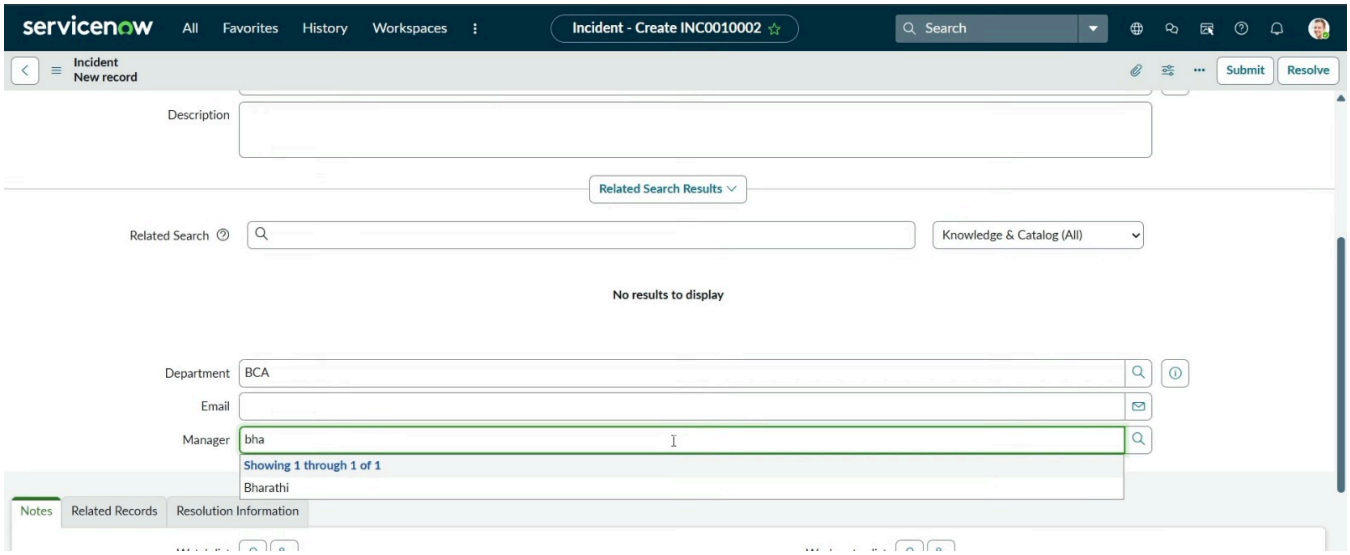

RESULT

- Employee data successfully imported from CSV into ServiceNow using Import Sets • Transform Map (employee transform) mapped all fields correctly to sys_user table • Processed: 4 inserts, 0 updates, 0 errors — clean import
- Dot Walking configured on Incident form — Department, Email, Manager displayed • Relationship mapping between employees, departments and managers working correctly
- No manual data entry required — full automation achieved

CONCLUSION

This project successfully demonstrates how to automate data import and relationship mapping in ServiceNow using Import Sets and Dot Walking. By loading employee data from a CSV file into a staging table (u_import_employee) and using a Transform Map to move it to the sys_user table, manual data entry was eliminated entirely.

Dot Walking was configured on the Incident form to automatically display related fields — Department, Email, and Manager — by traversing the relationship between the Incident and User tables. This improves data quality, reduces manual effort, and ensures accurate relationship mapping across ServiceNow tables.

This project demonstrates best practices in:

- Data Import Automation using Import Sets
- Field Mapping using Transform Maps and Mapping Assist
- Relationship Mapping using Dot Walking
- ServiceNow Form Configuration
- ServiceNow Administration and Development